

Md. Sakib Hasan

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EDUCATION

Daffodil International University

B.Sc. in Software Engineering

Dhaka, Bangladesh

2024 - 2028

Rajbari Government College

Higher Secondary Certificate (Science)

Rajbari, Bangladesh

2021 - 2023

TECHNICAL INITIATIVES & SYSTEMS DEVELOPMENT

Web Desktop Environment & System Architecture — Built SakibOS, a browser-based OS simulation with custom window management, drag-resize mechanics, reactive state engine, and dynamic UI rendering using CSS variable injection.

AI Systems & Multi-Model Orchestration — Developed a modular AI workspace integrating Ollama, OpenAI, and Groq with structured context pipelines, session memory, and SQLite-based persistent state management.

Computer Vision Systems — Implemented YOLOv8-based real-time object detection and MediaPipe-based gesture control system for low-latency interaction and tracking pipelines.

System Documentation & Cross-Domain Engineering — Created technical documentation across FinTech systems, AI workflows, and software architectures focusing on system design, transaction flow mapping, and interoperability models.

Web Graphics Engine — Built deterministic canvas rendering engine with AABB collision detection, optimized frame loop execution, and cross-platform rendering stability.

ENGINEERING EXPERIENCE

Designed modular software systems using API-driven architecture and local databases (SQLite, MongoDB). Managed scalable system workflows with state optimization, performance tuning, and structured Git-based development practices for collaborative engineering.

TECHNICAL SKILLS

Frontend: React, JavaScript, HTML, CSS

Backend: Node.js, Python, Express

Databases: MongoDB, MySQL, SQLite

Tools: Git, Docker, Linux, VS Code, GitHub Pages

Core: System Design, Full-Stack Development, AI Integration, Computer Vision, Technical Documentation